

RYAN ABBOTT

AI Engineer | Generative AI | Machine Learning Engineer

Oklahoma City, OK | Open to Dallas Hybrid / Remote | Top Secret Clearance
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AI/ML professional with 5+ years of experience designing, deploying, and scaling machine learning and AI systems across energy, cybersecurity, government, and scientific domains. Experienced in LLM fine-tuning, RAG architectures, distributed ML pipelines, MLOps, and cloud-native AI infrastructure on AWS and Databricks. Proven track record delivering multimillion-dollar operational savings, scalable inference services, and high-performance ML platforms supporting enterprise decision-making and automation.

CORE SKILLS

Generative AI / LLMs: RAG, MCP, Ontologies, Knowledge Graphs, Prompt Engineering, AI Agents, AI Orchestration

Machine Learning: Python, Scikit-learn, TensorFlow, PyTorch, Classification, Regression, NLP, LLM Fine-Tuning, Entity Resolution, Data Linkage, Forecasting, Optimization, Anomaly Detection

Cloud / MLOps: AWS, Databricks, Docker, GitHub Actions, CI/CD, EC2, ECR, S3, MLflow

Data Engineering: Spark, SQL, ETL, FastAPI, APIs, Lakehouse Architecture, Data Pipelines

Visualization / Tools: Power BI, Streamlit, React, Linux, Git

Languages: Python, SQL, C#, JavaScript, TypeScript, React, Kotlin, Java, C++, R

PROFESSIONAL EXPERIENCE

Independent Consulting — AI / ML Consultant

Oklahoma City, OK | Apr 2026 – Present

- Advised enterprise client on ontology architecture, knowledge graph design, and AI-ready data modeling for large-scale business systems.
- Delivered strategic recommendations for scalable knowledge base implementation supporting search, automation, and GenAI use cases.

Liberty Energy Inc. — AI / ML Software Engineer

El Reno, OK | Apr 2025 – Apr 2026

- Deployed production ML solution on Databricks achieving 96%+ accuracy and generating multimillion-dollar fuel-efficiency savings.
- Developed AI-driven optimization models to maximize fleet throughput while minimizing fuel consumption and operational cost.
- Led LLM fine-tuning initiative translating complex operational metrics into executive-friendly insights.
- Developed distributed Spark pipelines for telemetry analytics and large-scale model training.
- Designed predictive maintenance systems reducing downtime and preventing costly equipment failures.
- Built internal APIs on AWS using Docker, EC2, and ECR to support scalable AI applications.
- Implemented CI/CD workflows improving deployment speed, testing coverage, and release quality.
- Collaborated cross-functionally with engineering and operations teams to productionize AI solutions in field environments.

IntelliBridge, LLC — Data Scientist

Oklahoma City, OK | Jun 2023 – Apr 2025

- Built enterprise entity resolution platform using Python and FastAPI achieving 99% F1 score across person and company matching.
- Deployed containerized ML services to AWS for cross-team production usage.
- Developed anomaly detection models for cybersecurity log data to identify threats in near real time.
- Built Databricks lakehouse ingesting CloudWatch and CloudTrail telemetry for ML and analytics workloads.
- Delivered production data product supporting federal spending analysis and audit workflows.
- Built reusable validation frameworks improving data quality across multiple pipelines.
- Mentored junior analyst and led technical onboarding initiatives.
- Presented technical architecture to executives and engineering leadership.

University of Oklahoma — Research Assistant (Machine Learning)

Norman, OK | Aug 2022 – May 2023

- Designed TensorFlow CNN that doubled known rare quasar detections with 99%+ accuracy.
- Engineered terabyte-scale preprocessing pipelines for astronomical imaging data.
- Improved model performance through synthetic data generation and hyperparameter optimization.
- Presented findings to university faculty.

SELECTED AI PROJECTS

RyBot — RAG + MCP AI Assistant

Python | MCP | RAG | LLMs | Prompt Engineering

- Built AI-powered personal assistant built on RAG + Model Context Protocol that answers questions about me from structured personal knowledge sources.
- Developed conversational orchestration logic integrating external tools and memory-aware interactions.
- Deployed full-stack application with React-based UI and Python backend for intuitive, real-time interaction.

FeLoNet — Convolutional Neural Network

Python | TensorFlow | Neural Networks | Model Evaluation

- Developed a convolutional neural network for identifying rare quasars as part of academic research.
- Achieved 99%+ classification accuracy while significantly increasing known rare object detections.
- Optimized model performance through synthetic data generation and hyperparameter tuning.

EDUCATION

University of Oklahoma

B.S. Astrophysics, Minor in Mathematics

Graduated with Distinguished Honors | GPA: 3.62

CERTIFICATIONS

- Databricks Generative AI Fundamentals
- Databricks AI Security Fundamentals
- Databricks Data Governance Fundamentals
- Databricks Lakehouse Fundamentals